ASSESSMENT BRIEF (COURSEWORK 1)

Programme	BSc Computer Science/ BEng Software Engineering		
Module Title	Data Engineering		
Module Code	CMP-X304-0		
Module Level	6	**Assessment Type(s)**	**Report and Demo** (both mandatory to pass)
Word Length / Duration	1500 words	**% contribution to module mark**	50%
Deadline (date & time) for Submission	6/03/2026	**Format/Location of submission**	PDF/ Moodle
Assessment Feedback date:	3 weeks after the demo		

Learning Outcomes

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below.

By successfully completing this assessment, you will be able…

LO1: Evaluate concurrency and parallelism approaches to support data engineering solutions.

LO2: Evaluate simple strategies for executing a distributed query to select the strategy that minimizes the amount of data transfer.

LO3: Identify appropriate transaction boundaries in application programs.

LO4: Evaluate the time to retrieve information, when indices are used compared to when they are not used.

LO5: Analyse the requirements for building a data delivery pipeline.

Employability and Professional Skills

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skill(s) which is(are) critical in professional context and jobs.

1. Build and manage end-to-end data pipelines

Demonstrate the ability to collect, store, process, and retrieve data using industry-standard tools such as Python, MySQL, Apache NiFi, and MongoDB.

2. Apply analytical and problem-solving skills

Design solutions, debug issues, optimise code, and evaluate the effectiveness of SQL and NoSQL data storage approaches.

3. Communicate technical decisions clearly

LO6: Work as a member of a development team recognising the different roles within a team and different ways of organising teams.	Produce well-structured documentation, explain your rationale, and present your work effectively through written reports and demonstrations. **4. Demonstrate teamwork and professional practice** Participate in group-based demos, manage tasks responsibly, and reflect workplace collaboration and organisational skills.

Assessment Requirements

The final mark will come from two components (Both components are mandatory to pass this assignment)

1. Submission through Moodle.
2. Demonstration.

Everything must be submitted within the due date to be considered for a complete submission for this assessment. This assignment is **GROUP-BASED** and **only one member of the group will upload the document**. Demonstration will only be held after the work has been submitted through the specified link on Moodle. Each group **must consist of exactly four (4) students**. and **all four members must contribute** to the completion of all tasks (Tasks 1–4). Task allocation and internal coordination are the responsibility of the group.

Case Study: "TechReads Retail Analytics"

TechReads is a mid-sized UK company specialising in online sales analytics for technical books**.** Their clients include publishers, bookstores, and universities. TechReads is currently facing challenges because their data ingestion process is manual, inconsistent, and slow. They want to automate the entire pipeline—from data collection to storage, querying, and analytics.

You have been hired as a Junior Data Engineer in their Data Engineering Team. Your first project involves designing a complete data pipeline that will:

- **Scrape product information** from public e-commerce websites.
- **Store and query the extracted data** in a MySQL relational database.
- **Automate scheduled data ingestion** using Apache NiFi.
- **Integrate MongoDB** as a secondary NoSQL store for semi-structured analytics.

The company expects you to deliver a fully functioning prototype so they can evaluate your technical capabilities.

Project Background

TechReads wants to analyse trends in Data Engineering books sold online—pricing, authors, star ratings, and release years. Their goal is to supply publishers with insights such as:

- Which books are trending?
- Which authors consistently receive high ratings?
- How pricing varies across platforms.

To support this, they need a reliable and automated solution.

Your tasks as part of the TechReads Data Engineering Team are outlined below.

Task 1: Web Scraping

TechReads wants an automated script to collect book information daily.

Your role: Build a Python web scraper that extracts at least **15 Data Engineering books** from an e-commerce site.

At least **FIVE (5)** Data fields required (Preferable as below, but may vary):

- Title
- Author(s)
- Year of publication
- Star rating
- Price

You must:

- Download webpage(s) using the *Requests* library.
- Parse HTML using *BeautifulSoup*.
- Structure your extracted data using *Pandas*.
- Save the extracted data into a **CSV file** named *techreads_books.csv*.

In your report, discuss about your code and rationale behind it within 200 - 250 words for the task. You must provide the screenshots of each operation/execution of the code created by you with proper commenting.

Task 2: MySQL Database Pipeline

TechReads uses MySQL for its structured analytics dashboards.

Your role:

- Install and configure MySQL locally.
- Create a database called **techreads_db**.
- Import your CSV file into a MySQL table.
- Create a schema diagram (draw.io) representing your table.

- Run a SQL query that extracts **three selected columns**, then sort results by any one column (e.g., price or rating).

In your report, discuss about your code and rationale behind it within 200 - 250 words for the task. You must provide the screenshots of each operation/execution of the code created by you with proper commenting.

Task 3: Apache NiFi Dataflow Automation

TechReads wants to automate ingestion so that teams don't run scripts manually.

Your role:

- Create an Apache NiFi dataflow under a folder named "**DataEngineering**".
- Configure NiFi to pull book data from **your MySQL database**, transform it if needed, and save it to a **local directory** in a structured format (CSV, JSON, or Avro).
- Upload a video of the actual steps performed for carrying out the task **(with voiceover and camera ON)**

Task 4: MongoDB Integration & Performance Comparison

TechReads is exploring NoSQL databases to support flexible and scalable catalog analytics.

Your role:

- Convert your scraped data into JSON format.
- Insert the JSON dataset into a MongoDB database.
- Run a MongoDB query to filter books by attributes such as:
  - price,
  - publication year,
  - rating thresholds.
- Compare query performance between MySQL (SQL) and MongoDB (NoSQL) using similar query logic.

In your report, compare SQL database and NoSQL database and compare the query execution time for both the databases (within 200 - 250 words). You must provide the screenshots.

SUBMISSION REQUIREMENTS:

- The **cover page of the report must clearly list the names and student IDs of all four group members**.
- **Task 3 (Apache NiFi Video):**
  - Only **one group member** is required to record the NiFi demonstration video.
  - The video must include **voiceover and camera ON** while explaining the steps performed.
- **The report must also include reflection (Approx. 750 words)** (strictly your own reflection, not generated using any AI tool) discussing:
  - Each members specific contribution to the group work,
  - What they learnt from the task,
  - Challenges faced and how they were addressed.

- All submissions must adhere strictly to the stated **formatting, word limits, referencing style (IEEE), and file-naming conventions**.

Your report should contain:

The report should use 12-point font in Times New Roman, 1-inch margins, and double-spaced. The report should be properly paged, paragraphed, and sectioned, and must include the following sections in order in a (.pdf) file.

- Cover Page with names of all the members
- Table of Content
- Tasks:
  - Task 1: Discussion along with Necessary Screenshot(s)
  - Task 2: Discussion along with Necessary Screenshot(s)
  - Task 3: Video Upload
  - Task 4: Discussion along with Necessary Screenshot(s)
- Individual Reflection

References: Referencing any point in your discussion section(s). If you have used ideas from anywhere other than the lecture notes and tutorial examples (e.g., from a book, the Internet, or a fellow student) then include a reference showing where the code or ideas came from and label your code carefully to show which parts are your and which parts are borrowed.

Supporting File:

The supporting file you submit with your report should be an interactive python file (.ipynb format), containing code you implemented with proper comments.

Marking Criteria

Total Marks: 100

Task / Criteria	**Outstanding**	**Excellent**	**Good**	**Fair**	**Poor**
Task 1: Web Scraping & Data Extraction	Robust, accurate, and well-structured scraping solution. All required fields extracted correctly. Clear evidence of independent problem-solving and critical justification.	Strong scraping solution with minor limitations. Data mostly accurate and well-structured with clear reasoning.	Functional scraping solution with acceptable data quality. Limited depth of justification.	Basic or fragile scraping approach with missing or inconsistent data. Minimal justification.	Incomplete, incorrect, or non-functional scraping solution.
	20 marks	**17 marks**	**14 marks**	**10 marks**	**5 marks**
Task 2: MySQL Database Design & Querying	Well-designed relational schema with appropriate data types and constraints. Efficient SQL queries with	Sound schema design and correct queries with good justification. Minor inefficiencies present.	Adequate schema and queries but limited optimisation or justification.	Basic schema with design flaws or partially correct queries.	Incorrect or missing database setup and/or queries.

	strong analytical justification.				
	20 marks	**17 marks**	**14 marks**	**10 marks**	**5 marks**
Task 3: Apache NiFi Automation (Video)	Fully functional and well-organised NiFi workflow. Clear and confident explanation of processors, data flow, and design decisions.	Functional workflow with clear explanation. Minor omissions in depth or clarity.	Basic working flow with limited explanation of design decisions.	Partially working flow or unclear demonstration.	Non-functional or missing NiFi workflow/video.
	10 marks	**8 marks**	**6 marks**	**4 marks**	**2 marks**
Task 4: MongoDB Integration & SQL vs NoSQL Analysis	Accurate JSON conversion and MongoDB integration. Queries correct and meaningful. Critical, well-structured SQL vs NoSQL comparison applied to the scenario.	Correct MongoDB integration with a clear and relevant comparison. Some critical insight demonstrated.	Functional MongoDB usage with a descriptive but limited comparison.	Basic MongoDB task with superficial or weak comparison.	MongoDB task incomplete or comparison absent/incorrect.
	20 marks	**17 marks**	**14 marks**	**10 marks**	**5 marks**
Individual Reflection, Technical Understanding & Professional Communication during demonstration	Demonstrates comprehensive understanding of the full pipeline. Clearly explains own contribution with strong technical justification. Communicates confidently, responds accurately to questions, and shows high professionalism and engagement.	Demonstrates strong understanding of the pipeline and own role. Clear explanations with good justification. Responds well to most questions and maintains professional engagement.	Shows reasonable understanding of own task and its place in the pipeline. Provides adequate explanation with limited depth. Answers basic questions competently.	Limited understanding of pipeline and own role. Explanations lack clarity or depth. Struggles to justify decisions or answer questions effectively.	Demonstrates little or no understanding. Unable to explain contribution or respond meaningfully to questions. Limited or no engagement.
	30 marks	**25 marks**	**20 marks**	**15 marks**	**9 marks**

Assessment Success Guidance

To achieve high marks, students should demonstrate more than basic task completion. Strong submissions will show:

- Robust technical implementation – A fully functional end-to-end pipeline (Scraping → MySQL → NiFi → MongoDB) with accurate and well-structured outputs.
- Clear justification of decisions – Explain *why* specific tools, schema designs, and database approaches were chosen.

- Critical comparison – Provide thoughtful analysis of SQL vs NoSQL in terms of performance, scalability, and suitability for the scenario.
- Clean and professional presentation – Well-commented code, organised screenshots, clear diagrams, and a structured report.
- Independent initiative – Evidence of deeper exploration (e.g., indexing, performance measurement, improved data handling).
- Confident demo performance – Clear explanation of individual contribution and ability to answer questions effectively.

Use of Artificial Intelligence (AI)

Please read and apply:

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4a%20AI%20principles%20and%20guidance%20-for%20Senate.docx

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI_.docx

The assessment is designed so that the use of AI during the assessment is possible. You must acknowledge any use of AI and appropriately cite all AI generated outputs. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions.

**You can find the student guidance on the use of AI in the Library and Nest*

Referencing

References: Referencing any point in your discussion section(s). If you have used ideas from anywhere other than the lecture notes and tutorial examples (e.g., from a book, the Internet, or a fellow student) then include a reference showing where the code or ideas came from and label your code carefully to show which parts are your and which parts are borrowed.

You should include your reference list at the end of your report. You must ensure you use the appropriate IEEE format for your subject area.

Your referencing must use the IEEE referencing style IEEE Citation Guidelines.

It is highly recommended that you use reference management software such as RefWorks that is provided by the university. Your report should have as many references as is required.

It is your responsibility to ensure that you have actually read all the material you reference, and that the references provided in your report are legitimate and NOT AI-generated.

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the **Wellbeing team** first, who can support you. Further details can be found on the mitigating circumstances portal.

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 40%; if it is over 14 days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.

Submission Checklist

Before you submit, ask yourself:

Have I fully answered the assessment brief?

Have I met the word count and formatting requirements?

Is my referencing complete and accurate?

Have I declared any AI use honestly?

Have I proofread my work?

Am I submitting through the correct platform before the deadline?